

High Performance Mono Audio CODEC

FEATURES

System

- High performance and low power multi-bit delta-sigma audio ADC and DAC
- I²S/PCM master or slave serial data port
- 256/384Fs and USB 12/24 MHz audio system clocks
- I²C interface

Mono ADC

- 24-bit, 8 to 96 kHz sampling frequency
- 101 dB signal to noise ratio, -90 dB THD+N
- One pair of analog input with differential input option
- Low noise pre-amplifier
- Support analog and digital microphone

Mono Digital Class D

- 8 to 96 kHz sampling frequency
- 102 dB signal to noise ratio
- 1.25W@4Ω
- Over current and temperature protection
- Pop and click noise suppression

Low Power

- 1.8V to 3.3V operation
- mW playback and record
- Low standby current

APPLICATIONS

- CAT1
- Camera
- General Purpose

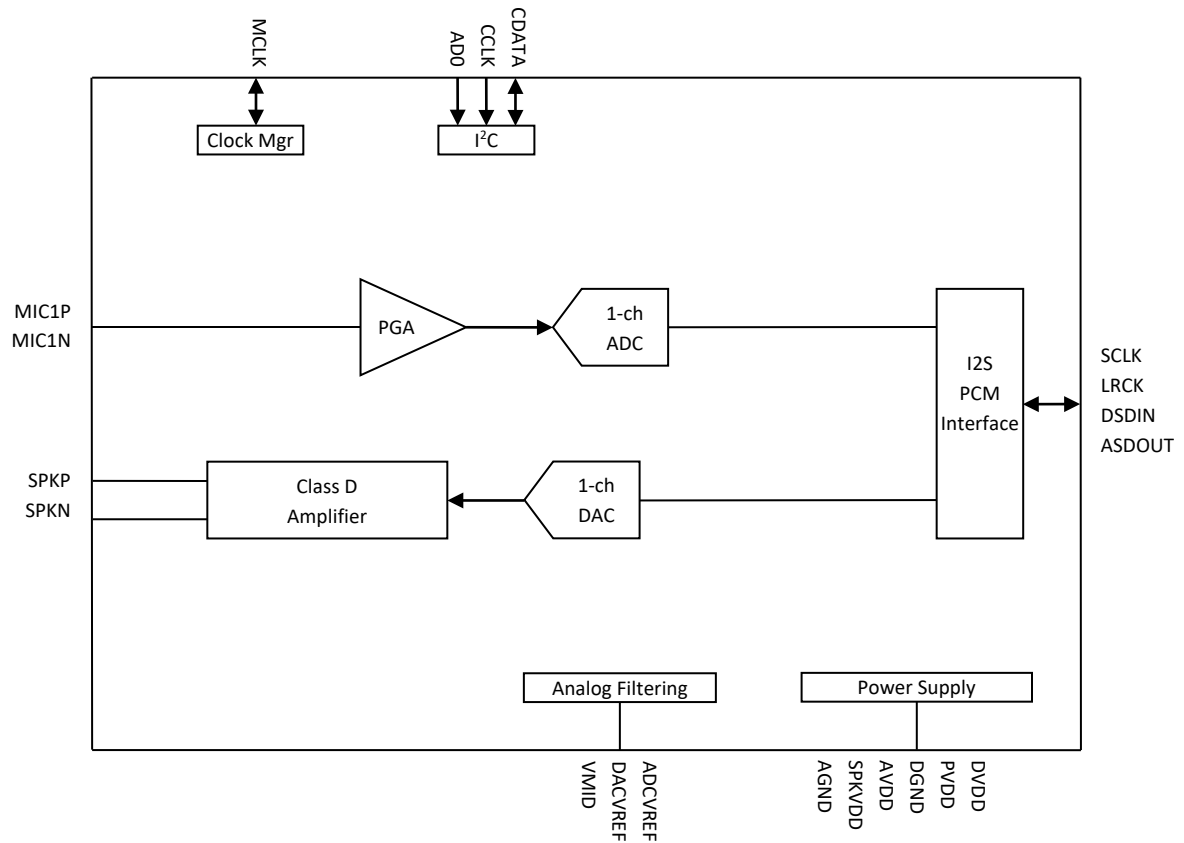
ORDERING INFORMATION

ES8375 -40°C ~ +125°C
3x3 QFN-20

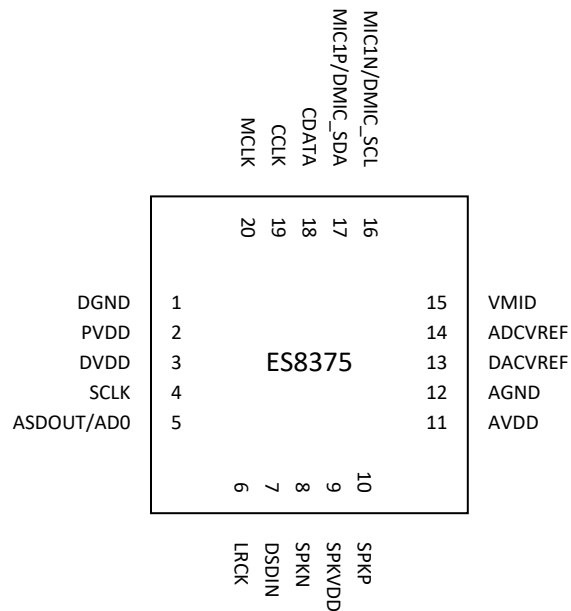
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1. BLOCK DIAGRAM

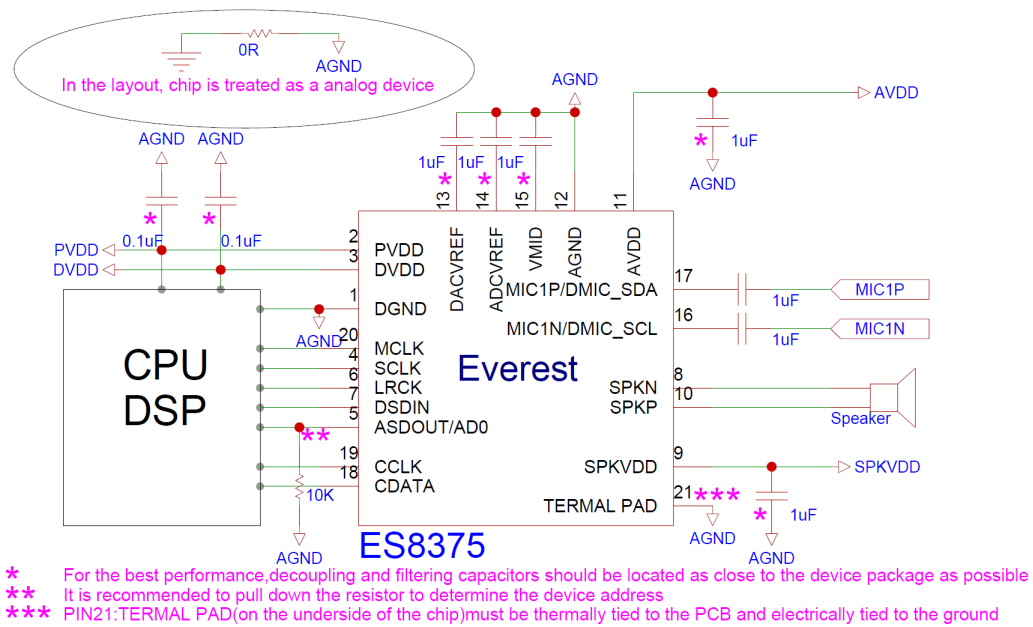


2. PIN OUT AND DESCRIPTION



Pin Name	Pin number	Input or Output	Pin Description
CCLK, CDATA	19, 18	I, I/O	I ² C clock, data
MCLK	20	I	Master clock
SCLK	4	I/O	Serial data bit clock
LRCK	6	I/O	Serial data left and right channel frame clock
ASDOUT/AD0	5	O/I	ADC serial data output/ I ² C address
DSDIN	7	I	DAC serial data input
MIC1P/DMIC_SDA MIC1N/DMIC_SCL	17 16	Analog/I Analog/O	Mic/line input or DMIC data and clock
SPKP, SPKN	10, 8	O	Mono speaker output
DVDD, PVDD, DGND	3, 2, 1	Analog	Digital power supply, PVDD for digital I/O
AVDD, AGND	11, 12	Analog	Analog power supply
SPKVDD	9	Analog	Speaker power supply
VMID	15	Analog	Filtering capacitor connection
ADCVREF, DACVREF	14, 13	Analog	Filtering capacitor connection
THERMAL PAD	21	Analog	Package bottom plate ground

3. TYPICAL APPLICATION CIRCUIT



4. CLOCK MODES AND SAMPLING FREQUENCIES

The device supports standard audio clocks (32Fs, 64Fs, 128Fs, 256Fs, 384Fs, 512Fs, etc.) and USB clocks (12/24 MHz).

The device can work either in master clock mode or slave clock mode. In slave mode, LRCK and SCLK are supplied externally, and LRCK and SCLK must be synchronously derived from the system clock with specific rates. In master mode, LRCK and SCLK are derived internally from device master clock.

5. MICRO-CONTROLLER CONFIGURATION INTERFACE

The device supports standard I²C micro-controller configuration interface. External micro-controller can completely configure the device through writing to internal configuration registers.

I²C interface is a bi-directional serial bus that uses a serial data line (CDATA) and a serial clock line (CCLK) for data transfer. The timing diagram for data transfer of this interface is given in Figure 1a and Figure 1b. Data are transmitted synchronously to CCLK clock on the CDATA line on a byte-by-byte basis. Each bit in a byte is sampled during CCLK high with MSB bit being transmitted firstly. Each transferred byte is followed by an acknowledge bit from receiver to pull the CDATA low. The transfer rate of this interface can be up to 400 kbps.

A master controller initiates the transmission by sending a “start” signal, which is defined as a high-to-low transition at CDATA while CCLK is high. The first byte transferred is the slave address. It is a seven-bit chip address followed by a RW bit. The chip address must be 001100x, where x equals AD0 (input pin: 1 being digital input high and 0 being digital input low). The RW bit indicates the slave data transfer direction. Once an acknowledge bit is received, the data transfer starts to proceed on a byte-by-byte basis in the direction specified by the RW bit. The master can terminate the communication by generating a “stop” signal, which is defined as a low-to-high transition at CDATA while CCLK is high.

In I²C interface mode, the registers can be written and read. The formats of “write” and “read” instructions are shown in Table 1 and Table 2. Please note that, to read data from a register, you must set R/W bit to 0 to access the register address and then set R/W to 1 to read data from the register.

Table 1 Write Data to Register in I²C Interface Mode

	Chip Address	R/W		Register Address		Data to be written		
start	001100 AD0	0	ACK	RAM	ACK	DATA	ACK	Stop

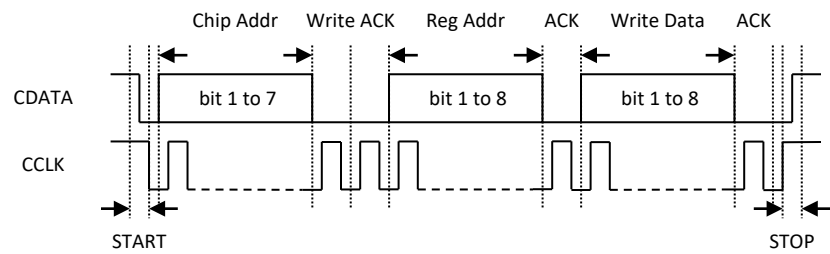


Figure 1a I²C Write Timing

Table 2 Read Data from Register in I²C Interface Mode

	Chip Address	R/W		Register Address		
Start	001100 AD0	0	ACK	RAM	ACK	
	Chip Address	R/W		Data to be read		
Start	001100 AD0	1	ACK	Data	NACK	Stop

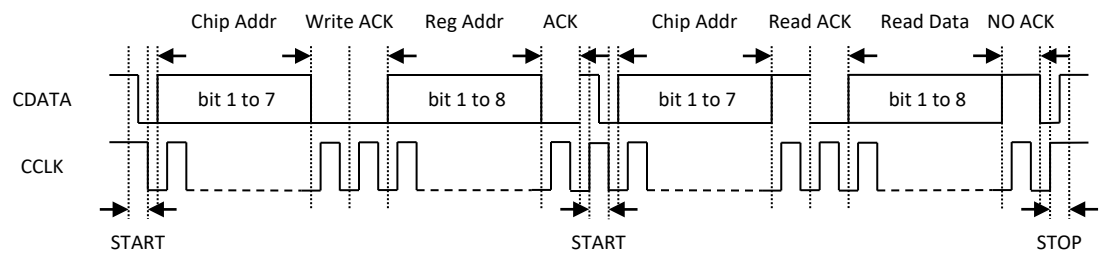


Figure 1b I²C Read Timing

6. I²S/PCM INTERFACE

The device provides many formats of serial audio data interface to the input of the DAC or output from the ADC through LRCK, SCLK and SDIN or SDOUT pins. These formats are I²S, left justified and DSP/PCM. DAC input SDIN is sampled by the device on the rising edge of SCLK. ADC data is out at SDOUT on the falling edge of SCLK. The relationship of SDATA (SDIN/SDOUT), SCLK and LRCK with these formats are shown through Figure 2a to Figure 2d.

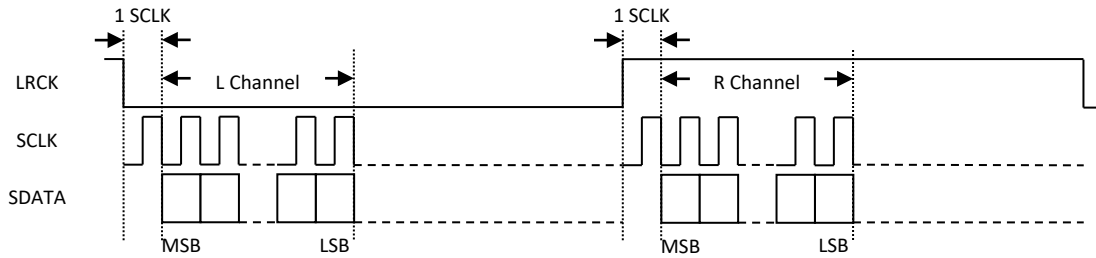


Figure 2a I²S Serial Audio Data Format

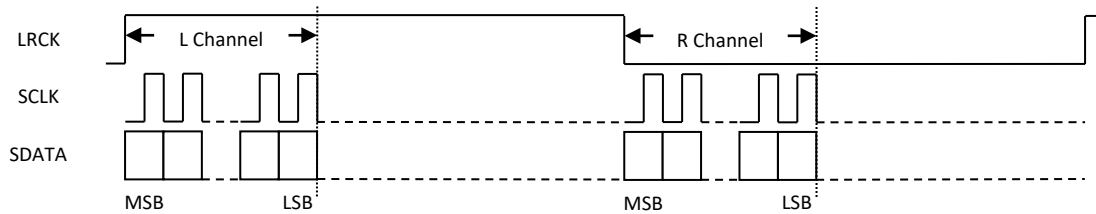


Figure 2b Left Justified Serial Audio Data Format

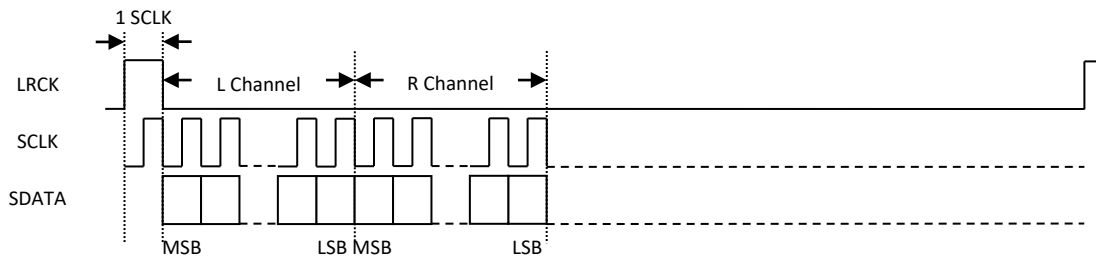


Figure 2c DSP/PCM Mode A Serial Audio Data Format

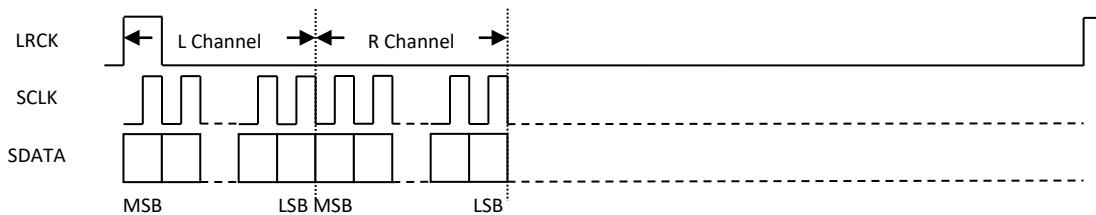


Figure 2d DSP/PCM Mode B Serial Audio Data Format

7. ELECTRICAL CHARACTERISTICS

ABSOLUTE MAXIMUM RATINGS

Continuous operation at or beyond these conditions may permanently damage the device.

PARAMETER	MIN	MAX
Analog Input Voltage Range	AGND-0.3V	AVDD+0.3V
Digital Input Voltage Range	DGND-0.3V	PVDD+0.3V
Operating Temperature Range	-40°C	+125°C
Storage Temperature	-65°C	+150°C

RECOMMENDED OPERATING CONDITIONS

PARAMETER	MIN	TYP	MAX	UNIT
DVDD	1.6	1.8/3.3	3.6	V
PVDD (digital I/O voltage)	1.6	1.8/3.3	3.6	V
AVDD	3.0	3.3	3.6	V
SPKVDD	3.0	3.3	3.6	V

ADC ANALOG AND FILTER CHARACTERISTICS AND SPECIFICATIONS

Test conditions are as the following unless otherwise specify: AVDD=3.3V, DVDD=1.8V, AGND=0V, DGND=0V, Ambient temperature=25°C, Fs=48 KHz, MCLK/LRCK=256.

PARAMETER	MIN	TYP	MAX	UNIT
ADC Performance				
Signal to Noise ratio (A-weight)	97	101	103	dB
THD+N	-93	-90	-87	dB
Interchannel Gain Mismatch		0.1	0.5	dB
Gain Error			±5	%
Filter Frequency Response				
Passband	0		0.4535	Fs
Stopband	0.5465			Fs
Passband Ripple			±0.05	dB
Stopband Attenuation	70			dB
Analog Input				
Full Scale Input (differential P and N)		2.2*AVDD/3.3		Vrms
Input Impedance		16		KΩ

DC CHARACTERISTICS

PARAMETER	MIN	TYP	MAX	UNIT
Normal Operation Mode				
DVDD=1.8V, AVDD=3.3V		49		mW
Power Down Mode (Note 1)				
DVDD=1.8V, AVDD=3.3V		0		uA
Digital Voltage Level				
Input High-level Voltage	0.7*PVDD			V
Input Low-level Voltage			0.5	V
Output High-level Voltage		PVDD		V
Output Low-level Voltage		0		V

Note 1: recommend all power supply on, entering low power through control register setting, then stopping input clock.

SERIAL AUDIO PORT SWITCHING SPECIFICATIONS

PARAMETER	Symbol	MIN	MAX	UNIT
MCLK frequency			49.2	MHz
MCLK duty cycle		40	60	%
LRCK frequency			96	KHz
LRCK duty cycle (Note 2)		40	60	%
SCLK frequency	VDDD=1.8V VDDD=3.3V		13 26	MHz
SCLK pulse width low	T_{SCLKL}	16		ns
SCLK Pulse width high	T_{SCLKH}	16		ns
SCLK falling to LRCK edge (master mode only)	T_{SLR}		5	ns
LRCK edge to SCLK rising (slave mode only)	T_{LSR}	10		ns
SCLK falling to ASDOUT valid	VDDD=1.8V VDDD=3.3V		31.5 12	ns
LRCK edge to ASDOUT valid (Note 3)	VDDD=1.8V VDDD=3.3V		26 10	ns
DSDIN valid to SCLK rising setup time	T_{SDIS}	10		ns
SCLK rising to DSDIN setup time	T_{SDIH}	10		ns

Note 2: one SCLK period of high time in DSP/PCM modes.

Note 3: only apply to MSB of Left Justified or DSP/PCM mode B.

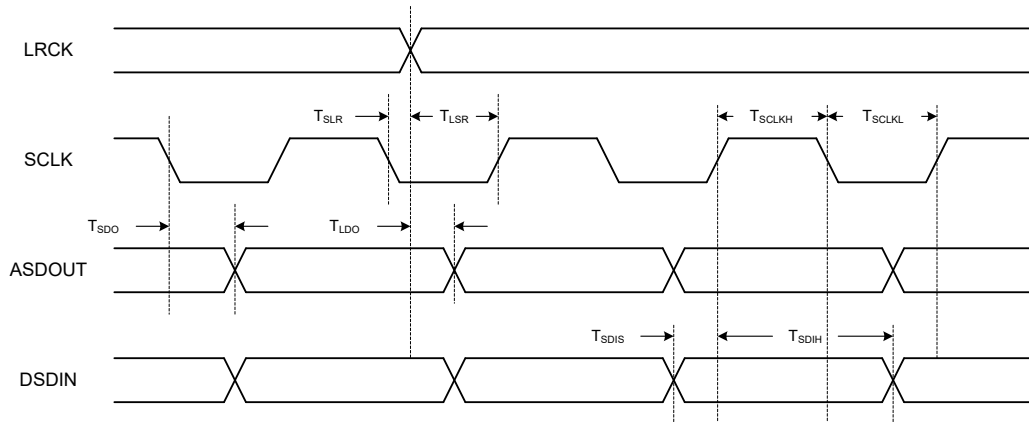
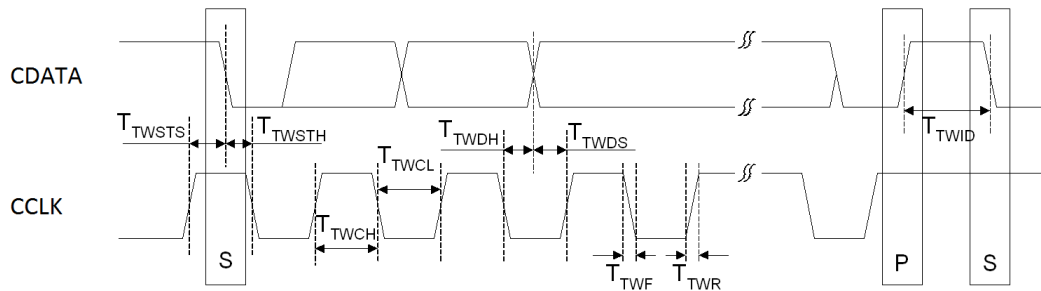


Figure 3 Serial Audio Port Timing

I²C SWITCHING SPECIFICATIONS (SLOW SPEED MODE/HIGH SPEED MODE)

PARAMETER	Symbol	MIN	MAX	UNIT
CCLK Clock Frequency	F _{CCLK}		100/400	KHz
Bus Free Time Between Transmissions	T _{TWID}	4.7/1.3		us
Start Condition Hold Time	T _{TWSTH}	4.0/0.6		us
Clock Low time	T _{TWCL}	4.7/1.3		us
Clock High Time	T _{TWCH}	4.0/0.6		us
Setup Time for Repeated Start Condition	T _{TWSTS}	4.7/0.6		us
CDATA Hold Time from CCLK Falling	T _{TWDH}		3.45/0.9	us
CDATA Setup time to CCLK Rising	T _{TWDS}	0.25/0.1		us
Rise Time of CCLK	T _{TWR}		1.0/0.3	us
Fall Time CCLK	T _{TWF}		1.0/0.3	us

Figure 4 I²C Timing

8. CONFIGURATION REGISTER DEFINITION

REGISTER 0X00 -RESET, DEFAULT 00111111

Bit Name	Bit	Description
MSC	7	Master/Slave select for SDP 0 - Slave mode 1 - Master mode
RST_REGS	6	Control port registers reset 0 - Not reset 1 - Reset
RST_DIG	5	CSM and SDP circuits reset 0 - Not reset 1 - Reset
RST_CMG	4	Clock manager circuit reset 0 - Not reset 1 - Reset
RST_MST	3	Master circuit reset 0 - Not reset 1 - Reset
RST_SPK	2	SPK circuit reset 0 - Not reset 1 - Reset
RST_ADC	1	ADC digital circuit reset 0 - Not reset 1 - Reset
RST_DAC	0	DAC digital circuit reset 0 - Not reset 1 - Reset

REGISTER 0X01 -RESET, DEFAULT 00000000

Bit Name	Bit	Description
RST_ADCDSP	6	ADC DSP circuit reset 0 - Not reset 1 - Reset
RST_ADCCF	5	ADC CombFilter circuit reset 0 - Not reset 1 - Reset
RST_ADCDEM	4	ADC DEM circuit reset 0 - Not reset 1 - Reset
RST_DACDSP	2	DAC DSP circuit reset 0 - Not reset 1 - Reset
RST_DACDSM	1	DAC DSM circuit reset 0 - Not reset 1 - Reset
RST_DACDEM	0	DAC DEM circuit reset 0 - Not reset 1 - Reset

REGISTER 0X02 -CLOCK MANAGER, DEFAULT 00000000

Bit Name	Bit	Description
MCLK_ON	7	MCLK in ON/OFF control 0 - MCLK off 1 - MCLK on
BCLK_ON	6	BCLK in ON/OFF control 0 - BCLK off 1 - BCLK on
ADC_ANACK_ON	5	ADC analog clock ON/OFF control 0 - off 1 - on
DAC_ANACK_ON	4	DAC analog clock ON/OFF control 0 - off 1 - on
ADC_DIGCLK_ON	3	ADC digital clock ON/OFF control 0 - off 1 - on
DAC_DIGCLK_ON	2	DAC digital clock ON/OFF control 0 - off 1 - on
SPKCLK_ON	1	Speaker clock ON/OFF control 0 - off 1 - on

REGISTER 0X03-CLOCK MANAGER, DEFAULT 00000000

Bit Name	Bit	Description
EXTCLK_SEL	7	External clock source select 0 - from MCLK PAD 1 - from BCLK PAD
MAINCLK_SEL	6	Internal clock "mainclk" source select 0 - from ext_clk(refer to EXTCLK_SEL) 1 - from osc_clk(Internal Oscillator clock)
MSTCLK_SEL	5	Master BCLK/LRCK divider clock source select 0 - from mainclk(refer to MAINCLK_SEL) 1 - from osc_clk(Internal Oscillator clock)
SPKCLK_SEL	4:3	Speaker clock source select 0 - from mainclk(refer to MAINCLK_SEL) 1 - from osc_clk(Internal Oscillator clock) 2 - from dac_anack(refer to DACCLK_SEL) 3 - not use
DMICCLK_SEL	2	DMIC clock source select 0 - from ADC OSR clock, align with the falling edge of adc_cclk(adc_cclk refer to ADC_CFCLK_SEL) 1 - from ADC OSR clock, align with the rising edge of adc_cclk
MAINCLK_INV	1	Internal clock "mainclk" invert 0 - normal 1 - invert
BITCLK_INV	0	Bit clock "bitclk" invert 0 - normal 1 - invert

REGISTER 0X04-CLOCK MANAGER, DEFAULT 00000000

Bit Name	Bit	Description
DIV_PRE	7:3	mainclk pre_divider clk_out = mainclk/(DIV_PRE+1) 0 - divide by 1
DLL_MODE	2:0	DLL mode control: 0 - DLL power down 1 - DLL bypass 2 - 2x 3 - 4x 4 - 6x 5 - 8x 6 - 12x 7 - 16x

REGISTER 0X05-CLOCK MANAGER, DEFAULT 00000000

Bit Name	Bit	Description
DMICCLK_DIV2	7	DMIC clock divide by 2 select 0 - not divide by 2 1 - divide by 2
ADCCLK_SEL	6:4	ADC analog clock select: 0 - dll_inclk 1 - dll_inclk x2 2 - dll_inclk x3 3 - dll_inclk x4 4 - dll_inclk x8 5 - dll_inclk/2 6 - dll_inclk/4 7 - dll_inclk/8
DACCLK_SEL	2:0	DAC analog clock select: 0 - dll_inclk 1 - dll_inclk x2 2 - dll_inclk x3 3 - dll_inclk x4 4 - dll_inclk x8 5 - dll_inclk/2 6 - dll_inclk/4 7 - dll_inclk/8

REGISTER 0X06-CLOCK MANAGER, DEFAULT 00000000

Bit Name	Bit	Description
ADC_CFCLKX2	7	ADC clock "adc_cfclk" x2 select 0 - x1 1 - x2
ADC_CFCLK_SEL	6:4	ADC clock "adc_cfclk" select 0 - dll_inclk 1 - dll_inclk x2 2 - dll_inclk x3 3 - dll_inclk x4 4 - dll_inclk x6 5 - dll_inclk x8

		6 - dll_inclk x12 7 - dll_inclk x16
DAC_DSMCLKX2	3	DAC clock “dac_dsmclk” x2 select 0 - x1 1 - x2
DAC_DSMCLK_SEL	2:0	DAC clock “dac_dsmclk” select 0 - dll_inclk 1 - dll_inclk x2 2 - dll_inclk x3 3 - dll_inclk x4 4 - dll_inclk x6 5 - dll_inclk x8 6 - dll_inclk x12 7 - dll_inclk x16

REGISTER 0X07-CLOCK MANAGER, DEFAULT 00000000

Bit Name	Bit	Description
ADC_DSPCLK_SEL	6:4	ADC clock “adc_dspclk” select 0 - dll_inclk 1 - dll_inclk x2 2 - dll_inclk x3 3 - dll_inclk x4 4 - dll_inclk x6 5 - dll_inclk x8 6 - dll_inclk x12 7 - dll_inclk x16
DAC_DSPCLK_SEL	2:0	DAC clock “dac_dspclk” select 0 - dll_inclk 1 - dll_inclk x2 2 - dll_inclk x3 3 - dll_inclk x4 4 - dll_inclk x6 5 - dll_inclk x8 6 - dll_inclk x12 7 - dll_inclk x16

REGISTER 0X08-CLOCK MANAGER, DEFAULT 00011111

Bit Name	Bit	Description
ADC_OSR	5:0	ADC OSR select 0~14 - not use 15 - 64*fs ... 31 - 128*fs ... 63 - 256*fs

REGISTER 0X09-CLOCK MANAGER, DEFAULT 00000000

Bit Name	Bit	Description
FLTCNT_DAC_HOLD	5:0	fltcnt_dac hold cycles, FLTCNT_DAC_HOLD = (dac_dspclk/LRCK - 256)/4 0 - 0 cycles

REGISTER 0X0A-CLOCK MANAGER, DEFAULT 00010101

Bit Name	Bit	Description
OSC_EN	7	Internal oscillator enable control 0 - power down 1 - enable
DIV_OSCCLK	6:4	osc_clk divider $\text{osc_clk} = \text{oscclk}/(\text{DIV_OSCCLK}+1)$ 1 – divide by 2
OSC_DET_CNT	2:0	Oscillator cycles for mclk off detect 0 - 2 osc cycles 1 - 4 osc cycles 2 - 8 osc cycles 3 - 16 osc cycles 4 - 32 osc cycles 5 - 64 osc cycles 6 - 128 osc cycles 7 - not use

REGISTER 0X0B-CLOCK MANAGER, DEFAULT 00000011

Bit Name	Bit	Description
DIV_BCLK	6:0	Master BCLK divider $\text{bclk_out} = \text{mstclk}/(\text{DIV_BCLK}+1)$ (mstclk refer to MSTCLK_SEL) 3 - divide by 4

REGISTER 0X0C-CLOCK MANAGER, DEFAULT 00100000

Bit Name	Bit	Description
CLKX2_PWSEL	7:5	ClockDoubler high pulse width select 0 – Level0 1 – Level4 2 – Level2 3 – Level6 4 – Level1 5 – Level5 6 – Level3 7 – Level7
DIV_LRCK[11:8]	3:0	Master LRCK divider bit 11 to bit 8 $\text{lrck_out} = \text{mstclk}/(\text{LRCK_DIV}+1)$ (mstclk refer to MSTCLK_SEL)

REGISTER 0X0D-CLOCK MANAGER, DEFAULT 11111111

Bit Name	Bit	Description
DIV_LRCK[7:0]	7:0	Master LRCK divider bit 7 to bit 0 $\text{lrck_out} = \text{mstclk}/(\text{LRCK_DIV}+1)$ (mstclk refer to MSTCLK_SEL) 255 – divide by 256

REGISTER 0X0E-CLOCK MANAGER, DEFAULT 00000000

Bit Name	Bit	Description
DIV_SPKCLK	6:0	Speaker clock divider $\text{spkclk} = \text{spkclk_src}/(\text{DIV_SPKCLK}+1)$ (spkclk_src refer to SPKCLK_SEL)

REGISTER 0X0F-CHIP STATE MACHINE, DEFAULT 00000000

Bit Name	Bit	Description
CSM_ON	7	Chip machine on/off 0 - csm off 1 - csm on
SEQ_DIS	6	Power up/down sequence 0 - Power up/down sequence enable 1 - Power up/down sequence power down
STDBY2NRML	5	STANDBY to NORMAL 0 - stay in STANDBY 1 - STANDBY to NORMAL
NRML2STDBY	4	NORMAL to STANDBY/ POWDOWN 0 - stay in NORMAL 1 - NORMAL to STANDBY/ POWDOWN
STDBY2PWDN	3	NORMAL to STANDBY/POWDOWN 0 - NORMAL to STANDBY 1 - NORMAL to POWDOWN
MCLKOFF_AUTO	2	NORMAL to STANDBY/ POWDOWN when mclk off 0 - stay NORMAL when mclk off 1 - NORMAL to STANDBY when mclk off
STDBY_LP	1	STANDBY low power mode 0 - normal power mode 1 - low power mode

REGISTER 0X15-SERIAL DATA PORT, DEFAULT 00000000

Bit Name	Bit	Description
DACDAT_SEL	7	DACDAT input select 0 - left channel 1 - right channel
DAC_S2P_MUTE	6	DAC s2p channel mute control 0 - unmute 1 - mute
SDP_LRP	5	SDP left right priority I2S/LJ case: 0 - L/R normal polarity Left/Right=High/Low (LJ) Left/Right=Low/High (I2S) 1 - L/R invert polarity Left/Right=Low/High (LJ) Left/Right=High/Low (I2S) DSP mode case: 0 - Mode A, MSB is available on 2nd BCLK rising edge after LRCK rising edge 1 - Mode B, MSB is available on 1st BCLK rising edge after LRCK rising edge
SDP_WL	4:2	SDP Wordlength select 0 - 24bit 1 - 20bit 2 - 18bit 3 - 16bit 4 - 32bit 5~7 - reversed
SDP_FMT	1:0	SDP format select

		0 - I2S 1 - LJ 2/3 - DSP
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REGISTER 0X16-SERIAL DATA PORT, DEFAULT 00000000

Bit Name	Bit	Description
ADCDAT_SEL	7:6	ADCDAT output select 0 - ADC + ADC 1 - ADC + 0 2 - 0 + ADC 3 - ADC + DAC
ADC_P2S_MUTE	5	ADC p2s channel mute control 0 - unmute 1 - mute

REGISTER 0X17-ADC DIGITAL CONTROL, DEFAULT 00000000

Bit Name	Bit	Description
ADC_SRC	7	ADC data source select 0 - AMIC 1 - DMICL
ADC_INV	6	ADC invert control 0 - Normal 1 - invert
ADC_RAMCLR	5	ADC RAM clear 0 - not clear 1 - clear ram
DMIC_POL	4	DMIC data polarity select 0 - Low=Left channel, High=Right channel 1 - Low=Right channel, High=Left channel
DMIC_GAIN	3:2	DMIC gain control: 0 - x1 1 - x2 2 - x4 3 - x8
ADC_LRCK_DLYSEL	1:0	adc_lrck delay select 0 - lrck delay1 1 - lrck delay2 2 - lrck delay3 3 - lrck delay4

REGISTER 0X18-ADC DIGITAL CONTROL, DEFAULT 00000000

Bit Name	Bit	Description
ADC_SYNC	7	ADC sync with LRCK 0 - not sync 1 - sync
ADC_RAMPRATE1	6:4	ADC volume softtrap rate1 0 - 0.125dB/LRCK 1 - 0.125dB/2LRCK 2 - 0.125dB/4LRCK 3 - 0.125dB/8LRCK 4~7 - disable softtrap

ADC_RAMPRATE2	3:0	ADC volume soframp rate2 0 - 0.125dB/LRCK 1 - 0.125dB/2LRCK 2 - 0.125dB/4LRCK 3 - 0.125dB/8LRCK 4 - 0.125dB/16LRCK 5 - 0.125dB/32LRCK 6 - 0.125dB/64LRCK 7 - 0.125dB/128LRCK 8~15 - disable soframp
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REGISTER 0X19-ADC DIGITAL CONTROL, DEFAULT 00011111

Bit Name	Bit	Description
ADC_OSR_GAIN	5:0	ADC OSR Gain 0 - '-31dB' 1 - '-30dB' ... 1dB/step 31 - '0dB' ... 62 - '31dB' 63 - '32dB'

REGISTER 0X1A-ADC DIGITAL CONTROL, DEFAULT 00000000

Bit Name	Bit	Description
ADC_VOLUME	7:0	ADC volume control 0x00 - '-95.5dB' 0x01 - '-95dB' ... 0.5dB/step 0xBE - '-0.5dB' 0xBF - '0dB' 0xC0 - '+0.5dB' ... 0xFF - '+32dB'

REGISTER 0X1D-ADC DIGITAL CONTROL, DEFAULT 00101100

Bit Name	Bit	Description
ADC_HPF	5	ADC offset freeze control 0 - freeze offset 1 - dynamic HPF
ADC_HPFS1	4:0	ADCHPF coefficient1 0x1F,0x0F is power down HPF 0x0C

REGISTER 0X1E-ADC DIGITAL CONTROL, DEFAULT 00001100

Bit Name	Bit	Description
ADC_HPFS2	4:0	ADCHPF coefficient2 0x1F,0x0F is power down HPF 0x0C

REGISTER 0X1F-DAC DIGITAL CONTROL, DEFAULT 00000100

Bit Name	Bit	Description
DAC_DSMMUTE	7	DAC DSM mute control 0 - not mute 1 - mute
DAC_DEMMUTE	6	DAC DEM mute control 0 - not mute 1 - mute
DAC_INV	5	DSP data phase control 0 - dsp data no phase inversion 1 - dsp data 180 degree phase inversion
DAC_RAMCLR	4	DAC RAM clear 0 - not clear 1 - clear ram
DAC_DITHER_OFF	3	DAC dsm dither on/off control 0 - on 1 - off
DAC_DSMCLIP	2	DSM clip when dsm overflow 0 - no clip 1 - clip
DAC_LRCK_DLYSEL	1:0	dac_lrck delay select 0 - lrck delay1 1 - lrck delay2 2 - lrck delay3 3 - lrck delay4

REGISTER 0X20-DAC DIGITAL CONTROL, DEFAULT 00000000

Bit Name	Bit	Description
DAC_SYNC	7	DAC sync with LRCK 0 - not sync 1 - sync
DAC_RAMPRATE1	6:4	DAC volume soframp rate1 0 - 0.125dB/LRCK 1 - 0.125dB/2LRCK 2 - 0.125dB/4LRCK 3 - 0.125dB/8LRCK 4~7 - disable soframp
DAC_RAMPRATE2	3:0	DAC volume soframp rate2 0 - 0.125dB/LRCK 1 - 0.125dB/2LRCK 2 - 0.125dB/4LRCK 3 - 0.125dB/8LRCK 4 - 0.125dB/16LRCK 5 - 0.125dB/32LRCK 6 - 0.125dB/64LRCK 7 - 0.125dB/128LRCK 8~15 - disable soframp

REGISTER 0X21-DAC DIGITAL CONTROL, DEFAULT 00000000

Bit Name	Bit	Description
DAC_VOLUME	7:0	DAC volume control

		0x00 - '-95.5dB' 0x01 - '-95dB' ... 0.5dB/step 0xBE - '-0.5dB' 0xBF - '0dB' 0xC0 - '+0.5dB' ... 0xFF - '+32dB'
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REGISTER 0X22-DAC DIGITAL CONTROL, DEFAULT 00011110

Bit Name	Bit	Description
DAC_VPPSCALE	5:0	DAC output scale 0 - '-3.875dB' 1 - '-3.75dB' ... 0.125dB/step 30 - '-0.125dB' 31 - '0dB' ... 62 - '+3.875dB' 63 - '+4dB'

REGISTER 0X27-ANALOG CONTROL, DEFAULT 00000000

Bit Name	Bit	Description
EN_OTP	7	Over Temperature Protection enable 0 - not enable 1 - enable
OTP_SEL	6:5	OTP level setting 0 - level1 1 - level2 2 - level3 3 - level4
OTP_WARN_LEVEL	4:3	DAC gain attenuation when OTP WARNING occur 0 - '0dB' 1 - '-3dB' 2 - '-6dB' 3 - '-12dB'
OTP_MASK	2	OTP WARNING FLAG mask control 0 - OTP FLAG ON 1 - MASK OTP FLAG

REGISTER 0X28- ANALOG CONTROL, DEFAULT 00000000

Bit Name	Bit	Description
EN_SPKBIASGEN	7	0 – disable SPK bias 1 – enable SPK bias
SPK_REG	6	0 – disable SPK 1 – enable SPK
SPK_PWM	5	0 – disable SPK MOD 1 – enable SPK MOD
SPK_HBDG	3	0 – disable SPK driver 1 – enable SPK driver
SPK_MUTEB	2	0 – mute SPK

		1 – unmute SPK
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REGISTER 0X29- ANALOG CONTROL, DEFAULT 00000000

Bit Name	Bit	Description
SPK_REF	6	0 – power down SPK REF 1 – enable SPK REF
LD2LSPK	5	0 – power down DAC to SPK 1 – enable DAC to SPK
SPK_LP	4	0 – disable SPK low power 1 – enable SPK low power
SPKL_DT	3	0 – disable SPK delay 1 – enable SPK delay
EN_SPKLSCP	2	0 – power down SPK short circuit protection 1 – enable SPK short circuit protection

REGISTER 0X2A- ANALOG CONTROL, DEFAULT 10101000

Bit Name	Bit	Description
SPKSEL1	7:6	SPK bias level 11 – high power 10 – normal 01 – low power 00 – off
SPKSEL2	5:4	SPK bias level 11 – high power 10 – normal 01 – low power 00 – off
SPKSEL3	3:2	SPK bias level 11 – high power 10 – normal 01 – low power 00 – off
SPK_TC	1:0	SPK clk selection 00 – 384k/400k 01 – 768k/800k 10/11 – do not use

REGISTER 0X2B- ANALOG CONTROL, DEFAULT 10111111

Bit Name	Bit	Description
SPKVOL	7:6	Speaker volume 0 - '-3dB' 1 - '0dB' 2 - '+3dB' 3 - '+6dB'

REGISTER 0X2E- ANALOG CONTROL, DEFAULT 01000000

Bit Name	Bit	Description
EN_ANA	7	0 – power down analog circuits 1 – enable analog circuits
MUTE_VREFQ	6	0 – mute VREFQ

		1 – unmute VREFQ
EN_IBIAS	5	0 – power down bias 1 – enable bias
EN_DACVREF	4	0 – power down DAC ref 1 – enable DAC ref
EN_DAC	3	0 – power down DAC 1 – enable DAC

REGISTER 0X2F- ANALOG CONTROL, DEFAULT 10001000

Bit Name	Bit	Description
VSEL	7:4	Internal setting 0100 – setting 4(lowest) 0101 – setting 5 0110 – setting 6 0111 – setting 7 1000 – setting 8 1001 – setting 9 1010 – setting 10 1011 – setting 11 1100 – setting 12 1101 – setting 13 1110 – setting 14 1111 – setting 15 (highest) Others – not allowed
VSEL1	3:0	Internal setting 0100 – setting 4(lowest) 0101 – setting 5 0110 – setting 6 0111 – setting 7 1000 – setting 8 1001 – setting 9 1010 – setting 10 1011 – setting 11 1100 – setting 12 1101 – setting 13 1110 – setting 14 1111 – setting 15 (highest) Others – not allowed

REGISTER 0X30- ANALOG CONTROL, DEFAULT 00001000

Bit Name	Bit	Description
VSEL2	3:0	Internal setting 0100 – setting 4(lowest) 0101 – setting 5 0110 – setting 6 0111 – setting 7 1000 – setting 8 1001 – setting 9 1010 – setting 10 1011 – setting 11 1100 – setting 12

		1101 – setting 13 1110 – setting 14 1111 – setting 15 (highest) Others – not allowed
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REGISTER 0X31- ANALOG CONTROL, DEFAULT 11000100

Bit Name	Bit	Description
BIAS_SW	7:6	0 – bias level0 1 – bias level1 2 – bias level2 3 – bias level3
DAC_SW	5	0 - Bias low 1 - Bias normal
VMIDSEL	3:2	0 – VMID=AVDD/2 1 – VMID lvl1 2 – VMID lvl2 3 – VMID lvl3
LP_DAC1	1	0 – normal mode 1 – low power mode
LP_DAC2	0	0 – normal mode 1 – low power mode

REGISTER 0X32- ANALOG CONTROL, DEFAULT 00001000

Bit Name	Bit	Description
EN_ADCBIASGEN	7	0 – power down 1 – enable
EN_ADCVREF	6	0 – power down 1 – enable
EN_PGA	5	0 – power down 1 – enable
EN_ADC	4	0 – power down 1 – enable
ADC_RST	3	0 – normal 1 – RESET

REGISTER 0X33- ANALOG CONTROL, DEFAULT 00000000

Bit Name	Bit	Description
LP_ADC0	7	0 – normal 1 – low power
LP_PGA	6	0 – normal 1 – low power
LP_PGA1	5	0 – normal 1 – low power
LP_ADC1	4:0	0 – normal 1 – low power

REGISTER 0X34- ANALOG CONTROL, DEFAULT 10001101

Bit Name	Bit	Description
ADC0_SW	7:4	0100 - setting level4 (lowest) 0101 - setting level5

		0110 - setting level6 0111 - setting level7 1000 - setting level8 1001 - setting level9 1010 - setting level10 1011 - setting level11 1100 - setting level12 1101 - setting level13 1110 - setting level14 1111 - setting level15 (highest) Others –not allowed
ADCBIAS_SW	3:2	00 - setting level1(lowest) 01 - setting level2 10 - setting level3 11 - setting level4
ADC1_SW	1:0	00 - setting level1(lowest) 01 - setting level2 10 - setting level3 11 - setting level4

REGISTER 0X35- ANALOG CONTROL, DEFAULT 10001000

Bit Name	Bit	Description
ADC2_SW	7:4	0100 - setting level4(lowest) 0101 - setting level5 0110 - setting level6 0111 - setting level7 1000 - setting level8 1001 - setting level9 1010 - setting level10 1011 - setting level11 1100 - setting level12 1101 - setting level13 1110 - setting level14 1111 - setting level15 (highest) Others –not allowed
ADC3_SW	3:0	0100 - setting level4(lowest) 0101 - setting level5 0110 - setting level6 0111 - setting level7 1000 - setting level8 1001 - setting level9 1010 - setting level10 1011 - setting level11 1100 - setting level12 1101 - setting level13 1110 - setting level14 1111 - setting level15 (highest) Others –not allowed

REGISTER 0X36- ANALOG CONTROL, DEFAULT 01010101

Bit Name	Bit	Description
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ADC4_SW	7:6	11 – high power 10 – normal 01 – low power 00 – off
ADC5_SW	5:4	11 – high power 10 – normal 01 – low power 00 – off
ADC6_SW	3:2	11 – high power 10 – normal 01 – low power 00 – off
ADC7_SW	1:0	11 – high power 10 – normal 01 – low power 00 – off

REGISTER 0X37- ANALOG CONTROL, DEFAULT 00000000

Bit Name	Bit	Description
LINSEL_GAIN	7	0 – normal 1 – add 6dB
LINSEL	6	0 – disable 1 – selection MIC input
PGAGAIN	3:0	0000 – 0dB 0001 – 3dB 0010 – 6dB 0011 – 9dB 0100 – 12dB 0101 – 15dB 0110 – 18dB 0111 – 21dB 1000 – 24dB 1001 – 27dB 1010 – 30dB Others – 0dB

REGISTER 0XF8- SYSTEM CONTROL, DEFAULT 00001111

Bit Name	Bit	Description
ADC2DAC_SEL	7	ADC data to DAC enable 0 - disable 1 - enable
TRI_BCLK	6	BCLK Tri-state control 0 - Normal 1 - Tri-state
TRI_LRCK	5	LRCK Tri-state control 0 - Normal 1 - Tri-state
TRI_ADCDAT	4	ADCDAT Tri-state control 0 - Normal 1 - Tri-state
PULLDOWN_ADCDAT	3	ADCDAT/CAD pulldown control

		0 - not pull down 1 - pull down Note: if there is no external pull-up or pull-down, do not turn off the internal pull-down
PULLDWN_DMICDATA	2	DMICDATA pulldown control 0 - not pull down 1 - pull down
PULLDWN_BCLK	1	BCLK pulldown control 0 - not pull down 1 - pull down
PULLDWN_LRCK	0	LRCK pulldown control 0 - not pull down 1 - pull down

REGISTER 0XF9- SYSTEM CONTROL, DEFAULT 00001100

Bit Name	Bit	Description
P2D_ISO_EN	3	PAD to digital signal isolate enable 0 - isolate power down 1 - isolate enable
A2D_ISO_EN	2	Analog to digital signal isolate enable 0 - isolate power down 1 - isolate enable
D2A_DET_EN	1	Analog detect digital power enable 0 - power down 1 - enable
A2D_DET_EN	0	Digital detect analog power enable 0 - power down 1 - enable

REGISTER 0XFA- SYSTEM CONTROL, DEFAULT 00000000

Bit Name	Bit	Description
PLAG_OTP	5	Over temperature flag, read only 0 - over temperature not occur 1 - over temperature occur
FLAG_SCP	4	Short Current flag, read write when short current occur auto set this bit to "1", when the fault clear, should write this bit to "0" to clear this flag, otherwise the csm can't startup
FLAG_MCLKOFF	2	MCLK OFF flag, read only 0 - MCLK on 1 - MCLK off

REGISTER 0XFD- SYSTEM CONTROL, DEFAULT 10000011

Bit Name	Bit	Description
CHIP_ID1	7:0	CHIP ID "1000 0011", read only

REGISTER 0XFE- SYSTEM CONTROL, DEFAULT 01110101

Bit Name	Bit	Description
CHIP_ID0	7:0	CHIP ID "0111 0101", read only

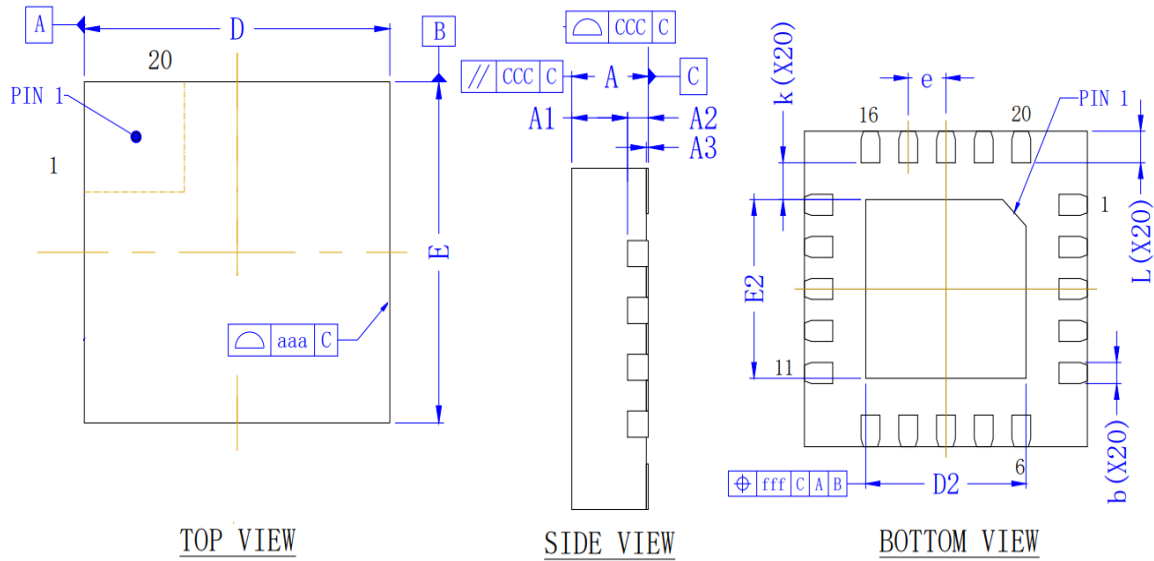
REGISTER 0XFF- SYSTEM CONTROL, DEFAULT 00000000

Bit Name	Bit	Description
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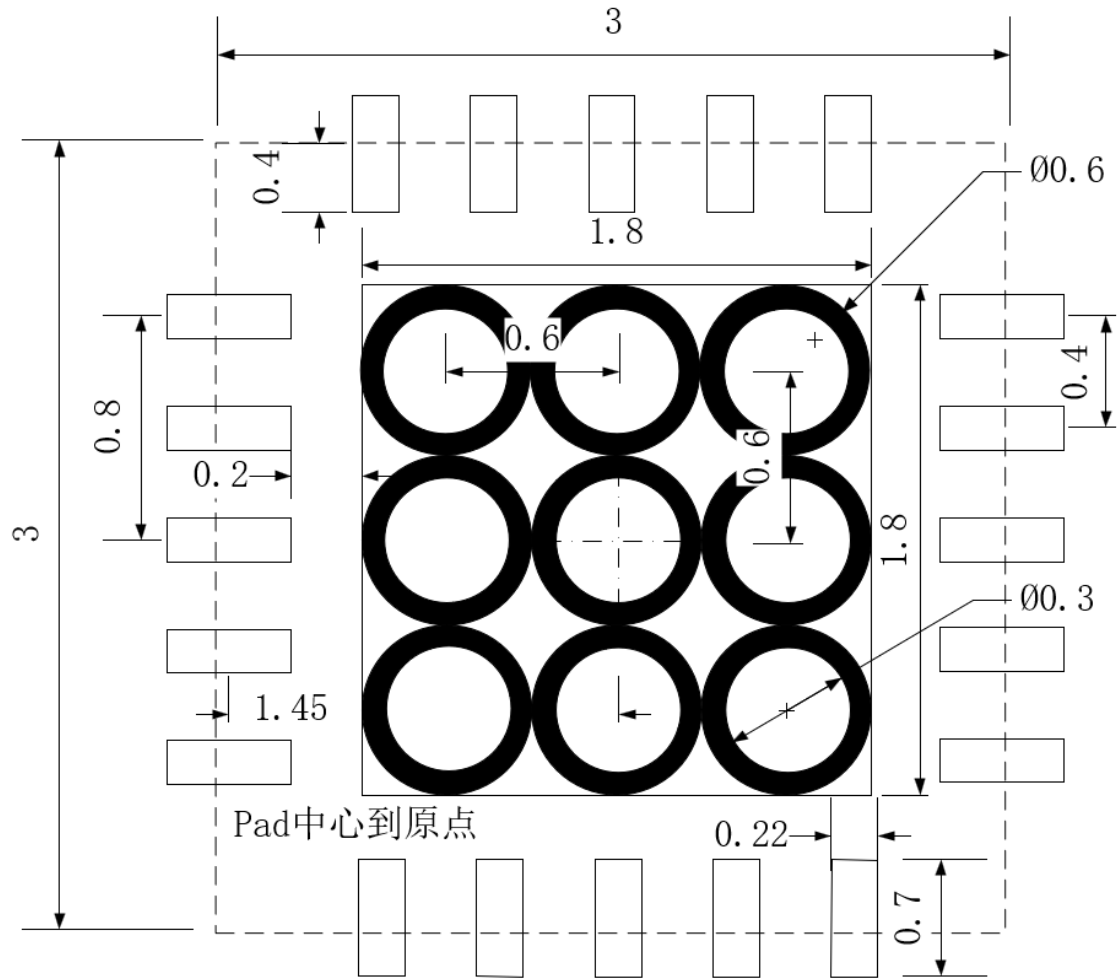
Security code	7:0	read only
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9. PACKAGE (UNIT: MM)

QFN-20L-3X3(P0.4T0.55) Package Outline Drawing



Item	Symbol	Minimum	Normal	Maximum
Total Thickness	A	0.50	0.55	0.60
Stand Off	A1	0.00	0.02	0.05
Molding Thickness	A2	---	0.40	---
LF Thickness	A3	0.152 REF		
Lead Width	b	0.15	0.20	0.25
Body Size	D	3 BSC		
	E	3 BSC		
Lead Pitch	e	0.4 BSC		
Exposed Pad Size	D2	1.60	1.70	1.80
	E2	1.60	1.70	1.80
Lead Length	L	0.20	0.30	0.40
Lead tip to Exposed Pad	K	0.35 REF		
Package Edge Tolerance	aaa	0.1		
Molding Flatness	ccc	0.1		
Coplanarity	eee	0.08		
Lead Offset	bbb	0.1		
Exposed Pad Offset	fff	0.1		



10. CORPORATE INFORMATION

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